

Introductory Project (1DT308)

Internet of Things (IoT) Basics

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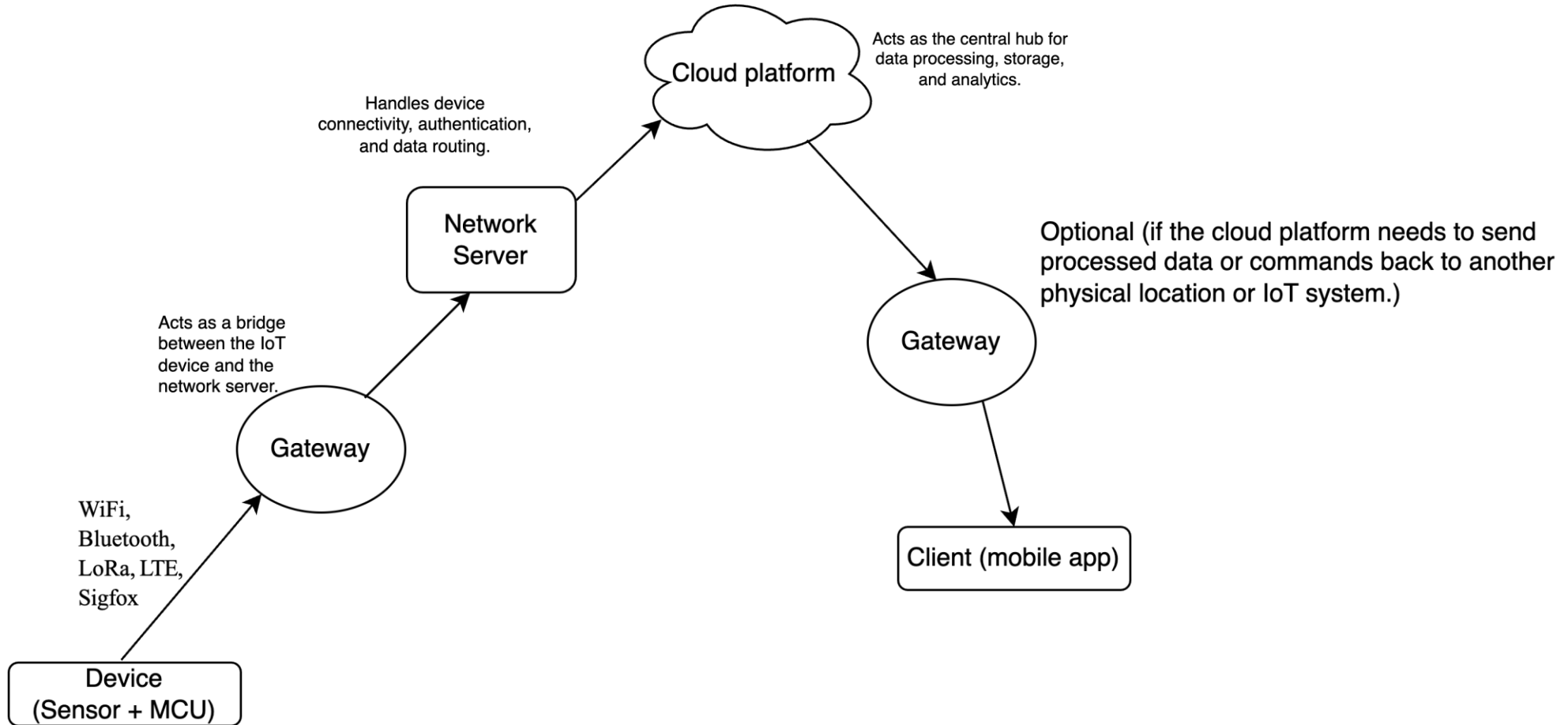
Faculty of Technology

Department of Computer science and Media Technology

Linnaeus University, Sweden.



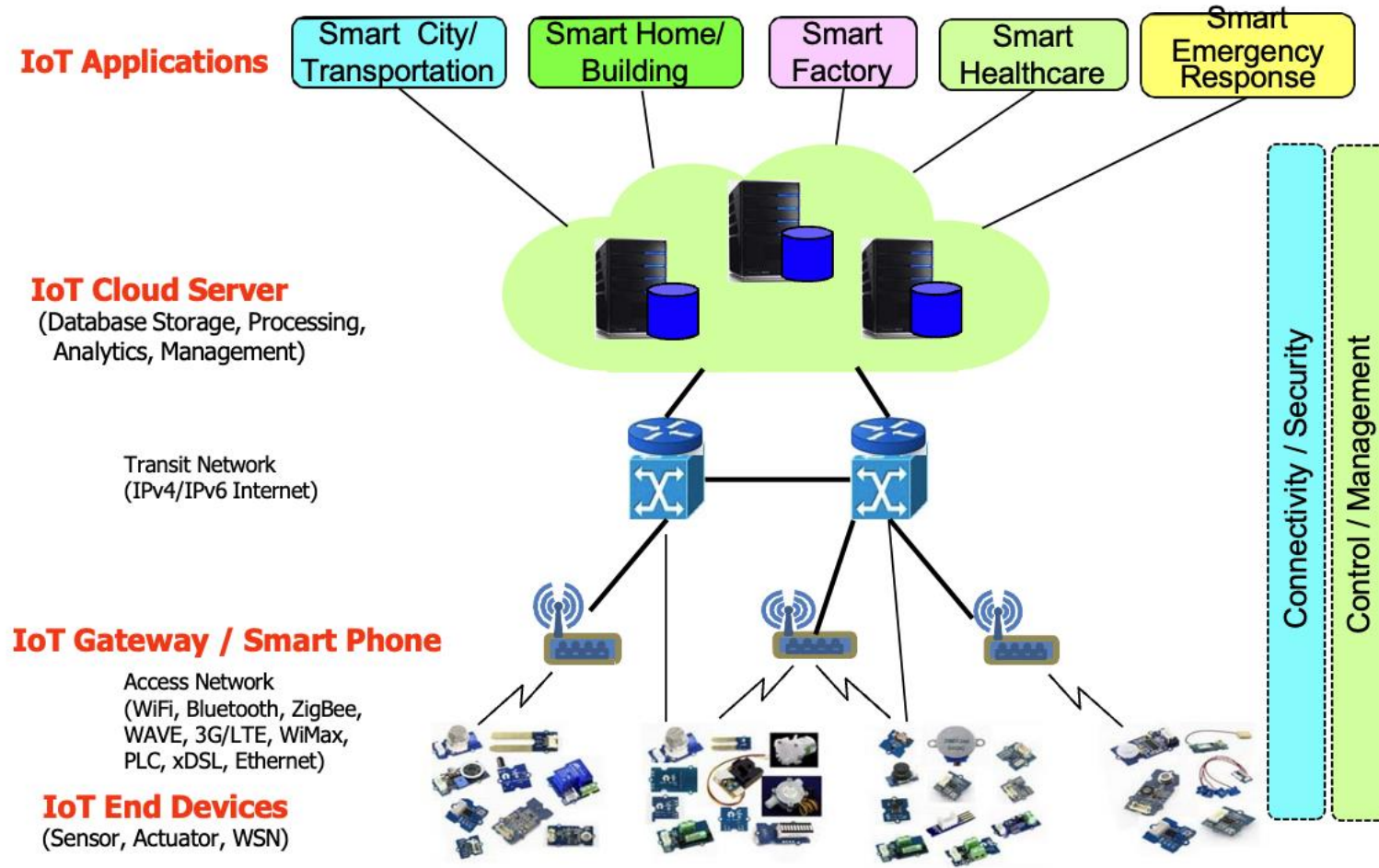
General Overview of IoT System



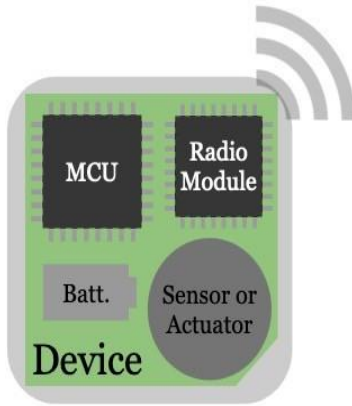
IoT Device → Gateway → Network Server → Cloud Platform (Backend & Frontend) → Mobile App



Conceptual View of IoT



Device



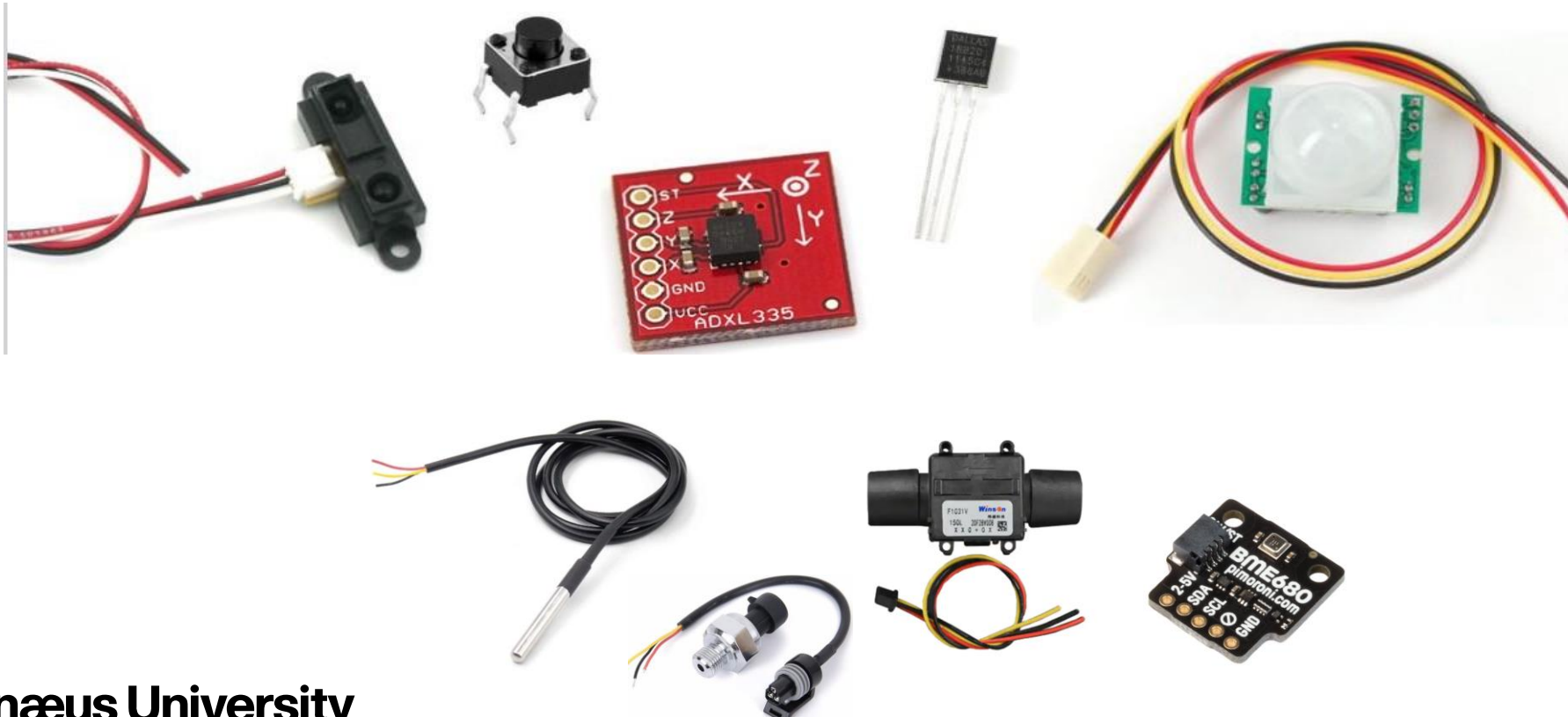
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- Embedded computer with **sensors and actuators**.
- **Connectivity** on the chip or as an external module.
- **Microcontroller (MCU)** with constrained resources.
- Small, slow processor, limited memory, low power.
- Often **battery-powered** or even harvesting energy.



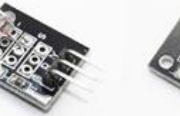
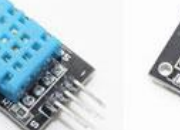
Sensors

- A sensor is a device that **detects and measures changes** in its environment such as temperature, pressure, light, or motion, and converts these changes (analog signals) into electrical signals.
- E.g. temperature, sound, light, distance, and motion.



Common IoT and Electronics Prototyping Sensors

Ultrasonic module 	Human body sensor module 	Tilt sensor 	Photosensitive sensor 
Smoke sensor 	Infrared barrier sensor 	Vibration sensor 	Sound sensor 
1 path search sensor 	Flame sensor 	Laser Head Sensor 	Clock module 
Super regenerative module 	Temperature and humidity sensor 	Raindrop sensor 	Soil Sensors 

JoyStick 	Flame 	RGB LED 	Heartbeat 	Light Cup 	Hall magnetic 
Relay 	Linear Hall 	SMD RGB 	7 color flash 	Tilt switch 	TEMP 
Big sound 	Touch 	Two-color 	Laser emit 	Ball switch 	Analog temp 
Small sound 	Digital temp 	Two-color 	Button 	Photoresistor 	IR emission 
Tracking 	buzzer 	Reed switch 	Shock 	Temp and humidity 	IR recelver 
Avoid 	Passive buzzer 	Mini Reed 	Rotary encoders 	Analog Hall 	Tap module Light blocking 

Commonly used in educational projects, prototyping, and hobbyist applications



Commercial Sensors



LHT65 LoRaWAN Temperature & Humidity Sensor



Wireless Infrared Object Detector Sensor 868MHz / 915MHz - LoRa



tabs LoRaWAN Temperature & Humidity Sensor TBHH100-868
Antratek Electronics

LoRaWAN Temperature & Humidity Sensor



Water Leak Sensor Strip



Wisblock Kit 1 - Indoor Ambient Monitor

By RAKwireless Technology Co.

The WisBlock Kit 1 is composed of RAK4631 WisBlock Core, RAK1901 Temperature and Relative Humidity, RAK1902 Barometric Pressure and RAK1903 Light Sensors.



LDDS75

By Dragino Technology Co., Limited

The Dragino LDDS75 consists of an ultrasonic detection sensor for measuring the distance between the sensor and a flat object. It is used to measure the distance between the sensor and a flat object. It uses temperature compensation to improve the reliability of data. The measured distance data are sent to a LoRaWAN® network for further analysis.



Parking Lot Sensor

By Bosch Connected Devices and Solutions GmbH

The BOSCH wireless smart parking sensor is a LoRaWAN® end device that uses a magnetometer and radar which are two independent sensors for parking space occupancy. It enables active parking lot management features, such as search, navigation, and reservation. In addition to that, the BOSCH parking lot sensor has a GPS receiver and a temperature sensor.



- Temperature
- Humidity
- Air Pressure
- Motion



Actuators

- An actuator in IoT is a device that performs **physical actions or movements** in response to external signals or commands, typically from a control system or sensor data.
- Actuators help IoT systems interact with the physical world by **converting electrical or digital signals into actions** (analog signals) like opening or closing a valve, moving a motor, or turning a switch.



Servo Motor



LED



Relay



Step Motor

Gateway

- Connects local devices/networks to the Internet.
- Can be transparent (forward data without modification) or convert payloads (data modification for interpreting the data).



WiFi gateway



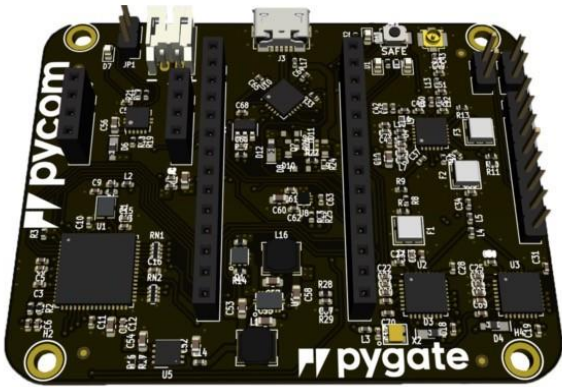
LoRaWAN Outdoor
gateway



LoRaWAN Indoor
gateway

Pygate Gateway with Europa (868 MHz) and EEUU (915 MHz) Frequency

Pygate 8-channel LoRaWAN gateway

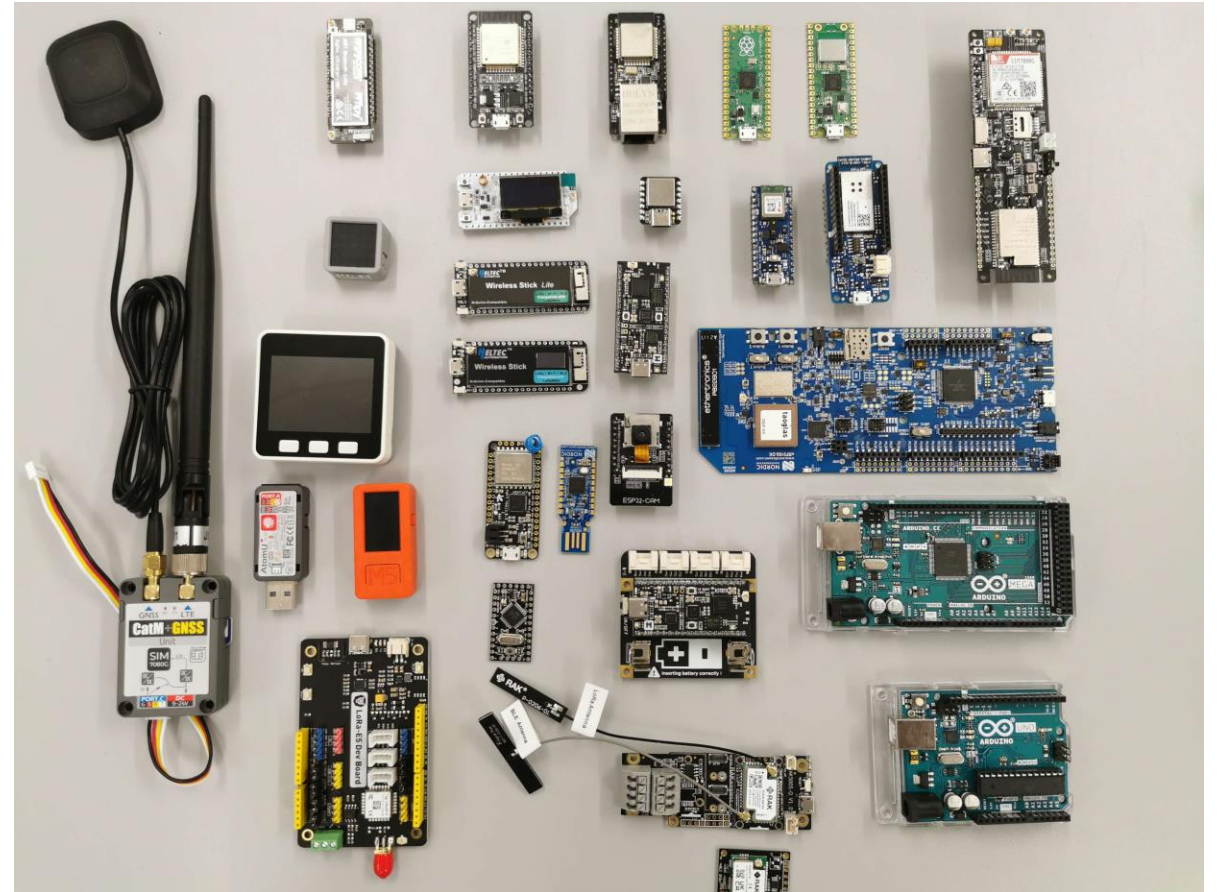


Pygate is an 8-channel LoRaWAN gateway. Connect your WiPy 3.0, LoPy4 and GPy and connect up any other Pycom devices for a decentralised IoT network.



Different Development Boards (Microcontrollers)

- Arduino boards
- Raspberry Pi boards
- ESP8266/ESP32 modules
- Particle Photon or similar microcontroller boards
- STM32 development boards
- Feather boards (Adafruit)
- Micro:bit
- Teensy boards
- BeagleBone boards
- NodeMCU

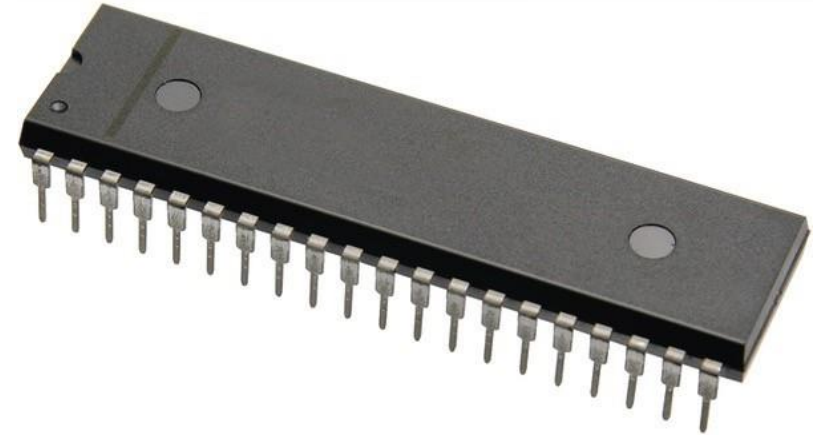




Arduino Uno



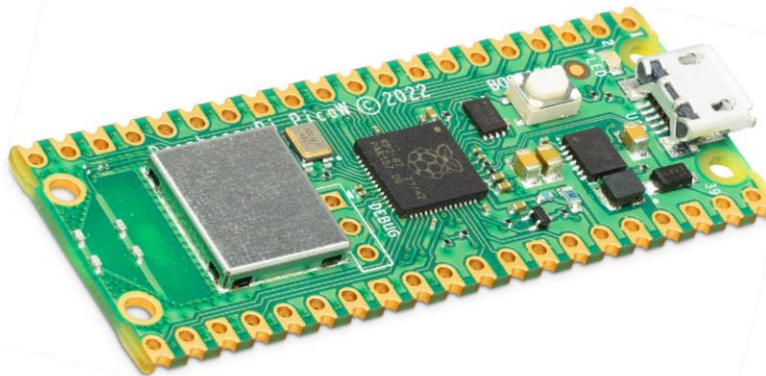
STM32F4



PIC16F877A



Raspberry Pi Pico



Raspberry Pi Pico W



ESP32



Development Boards: MCU vs. SBC

- Microcontrollers (MCU) are small computers that run a single program.
- An MCU has limited memory and a slow processor.
- An MCU has no operating system, i.e. less overhead.
- An MCU can react faster, in real time.
- Use microcontrollers for simple, low-latency tasks.
- Single-board computers (SBC) are not microcontrollers.
- SBC runs a full Linux OS, has a lot of memory, and uses way more power.

✓ MCU examples: Arduino, Raspberry Pi Pico, Raspberry Pi Pico W,...

✓ SBC examples: Raspberry Pi



Two Common Microcontrollers Designers/Developers

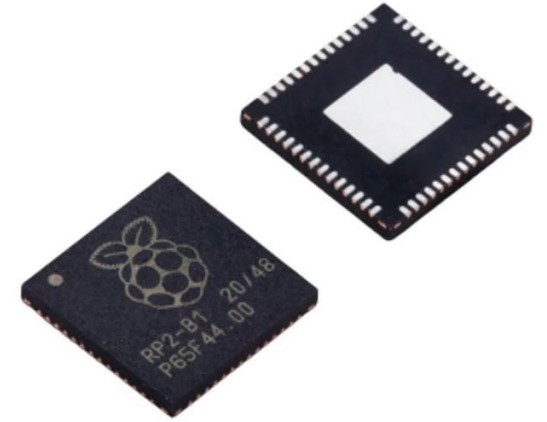
- Espressif systems
(Esp32, esp8266,...)
- Raspberry Pi Foundation
(RP2040, RP Zero, RP3, RP4, ...)



RP Pico W is an RP2040-based microcontroller announced in October 2021.



RP2040 Chip

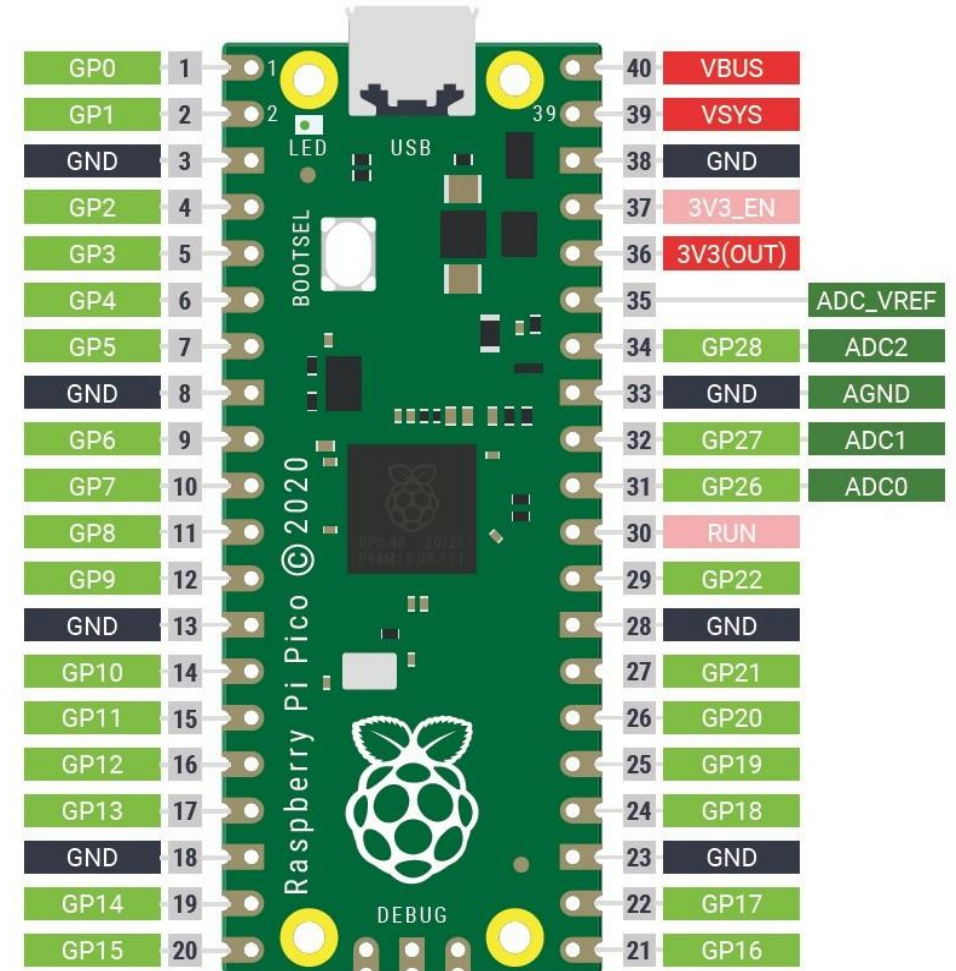


- **RP** means 'Raspberry Pi', simply enough.
- **2** is the number of *processor cores* the microcontroller has.
- **0** is the type of processor core, indicating in this case the RP2040 uses a processor core called the *Cortex-M0+* from Cambridge-based Arm.
- **4** is how much *random access memory (RAM)* the microcontroller has, based on a special mathematical function: $\text{floor}(\log_2(\text{RAM}/16))$. In this case, '4' means the chip has 264 kilobytes (kB) of RAM.
- **0** is how much *non-volatile (NV) storage* the chip has, and is worked out in the same way as the RAM: $\text{floor}(\log_2(\text{NV}/16))$. In this case, 0 simply means there is no non-volatile storage on-board.



The Raspberry Pi Pico W Pins, Seen from the Top of the Board

- Rpi Pico W talks to hardware through a series of pins along both its edges => GPIO
- Rpi Pico W has 40 pins labelled on the underside of the board, with three labeled with their numbers on the top of the board: Pin 1, Pin 2, and Pin 39.

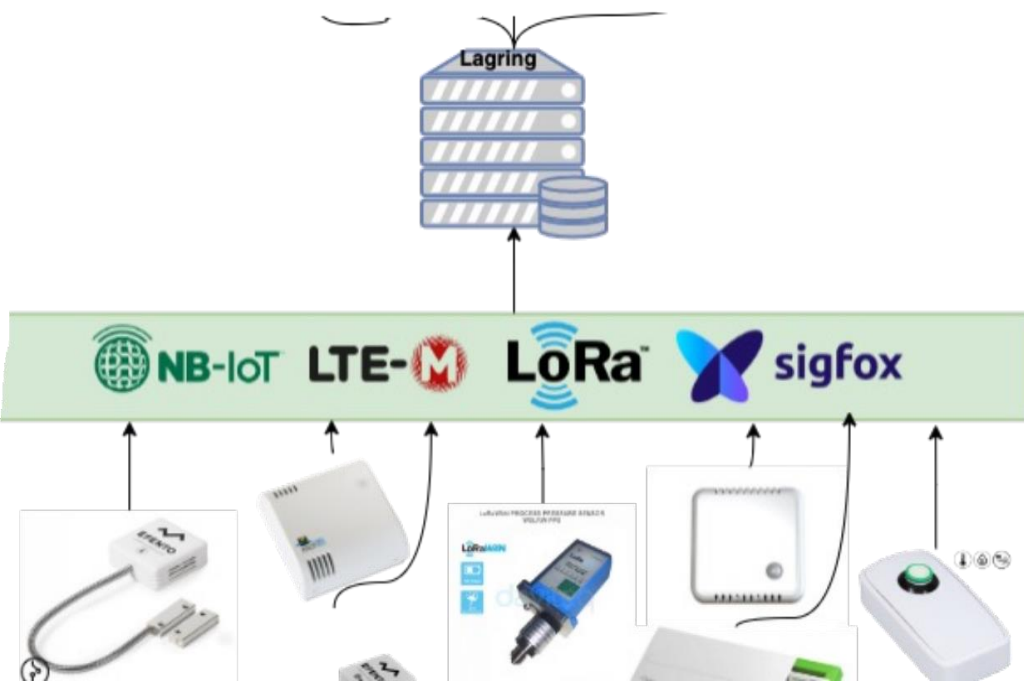


Note the mapping between the pin number to the GPIO number in your code.



Connectivity Options

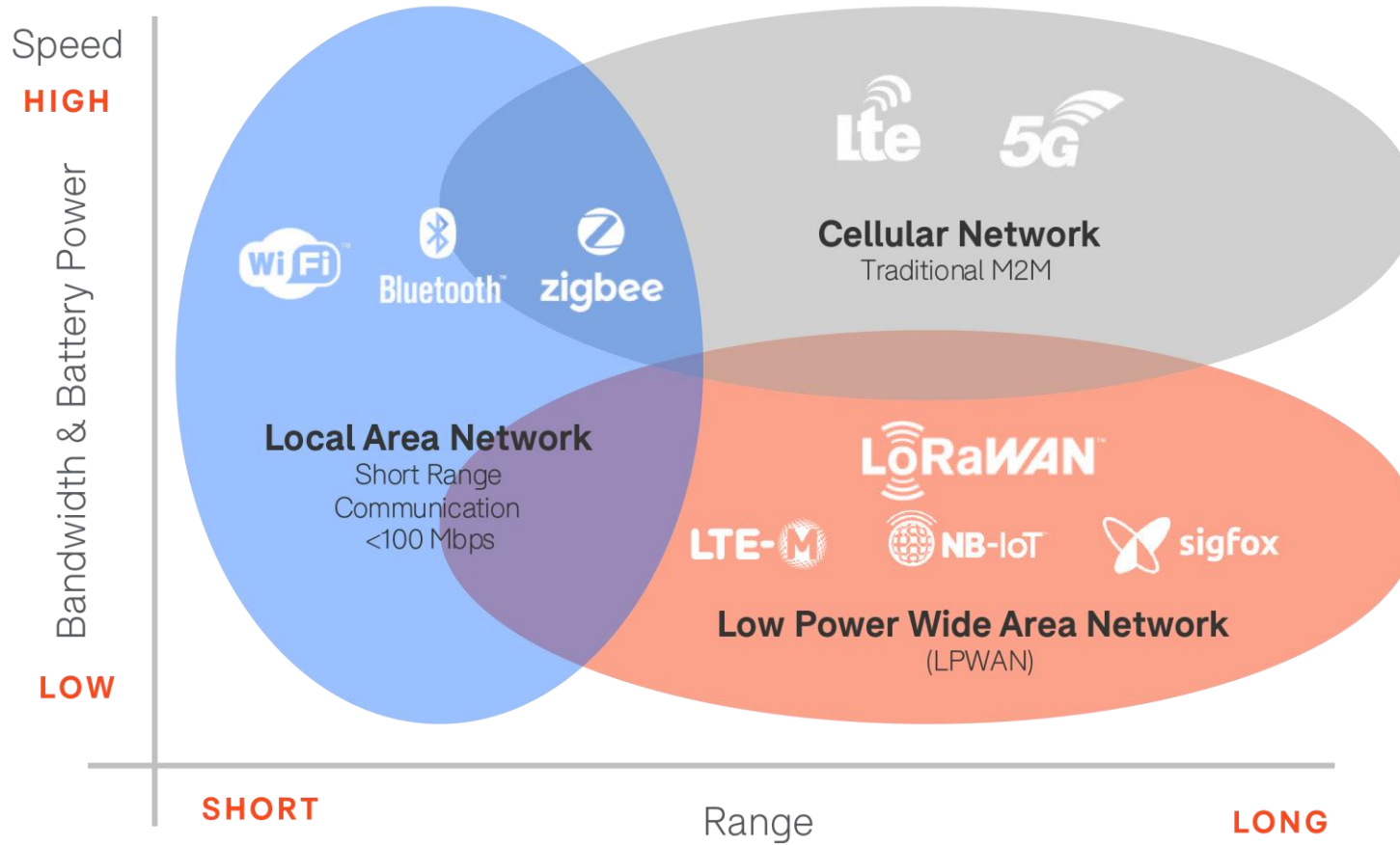
Technology	Description
Wi-Fi 6/6E (IEEE 802.11ax)	Fast, efficient Wi-Fi with improved range for IoT.
Wi-Fi 7 (IEEE 802.11be)	Next-gen Wi-Fi offering higher speeds and lower latency.
5G NR (New Radio)	High-speed, low-latency cellular for IoT with dense device support.
LoRaWAN	Low-power, long-range network for IoT.
Zigbee 3.0	Low-power, secure mesh network for IoT.
Thread	Low-power, IPv6 mesh networking for smart devices.
Bluetooth 5.x	Long-range, high-speed Bluetooth for IoT devices.
NB-IoT (Narrowband IoT)	Cellular tech for low-power, wide-area IoT applications.
Wi-SUN	Scalable network for smart utilities.
6LoWPAN	IPv6 protocol for low-power wireless IoT networks.



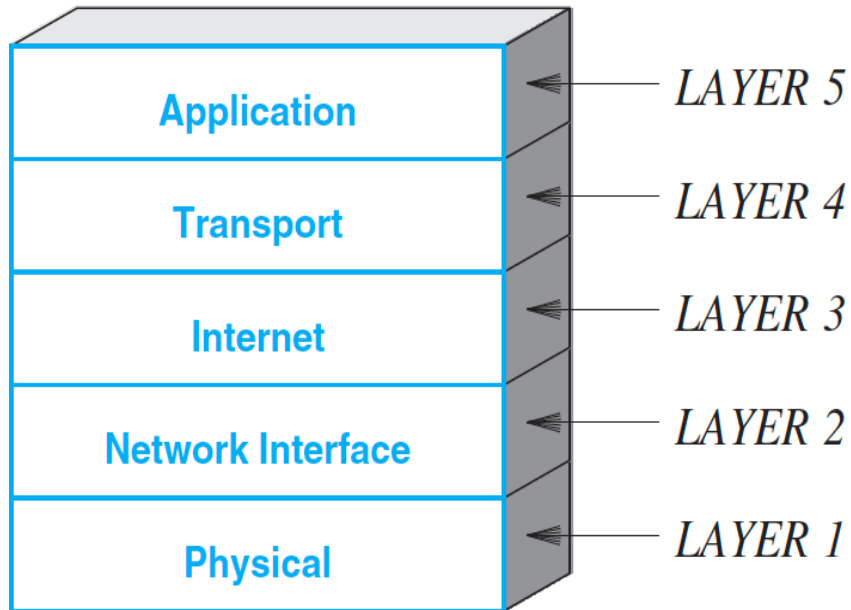
WiFi, Bluetooth, LoRa, LTE, Sigfox



Connectivity Options



Layering Model (TCP/IP)



The layering model used with the Internet protocols (TCP/IP)

Layer	TCP/IP (Web) Stack	IoT Network Stack
Application	HTTP/HTTPS, SMTP, FTP, DNS	CoAP, MQTT, AMQP
Transport	TCP, UDP	UDP, sometimes TCP
Network	IP	6LoWPAN, RPL
Physical & Data Link	Ethernet, Wi-Fi, IE	LoRa, Zigbee, BLE, IEEE 802.15.4



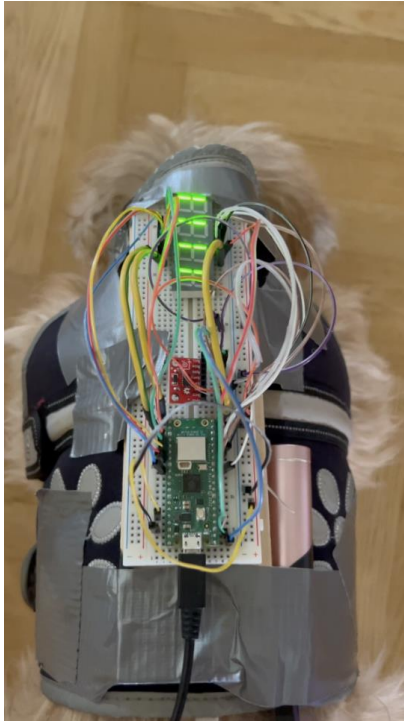
Example Projects

IoT Applications

- **Smart Home Automation:** Energy management, lighting, and security
- **Healthcare:** Remote patient monitoring, fitness tracking
- **Agriculture:** Precision farming, environmental monitoring
- **Transportation:** Fleet management, autonomous vehicles
- **Industry:** Predictive maintenance, smart manufacturing
- **Retail:** Smart inventory management, customer analytics
- **Environment:** Weather monitoring, pollution control
- **Education:** Smart classrooms, remote learning systems

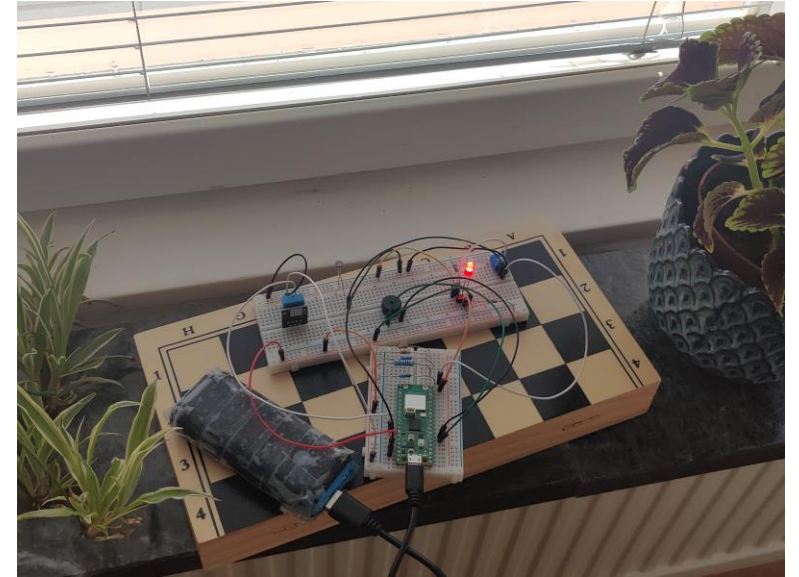


Examples from Summer 2024: Applied IoT Course

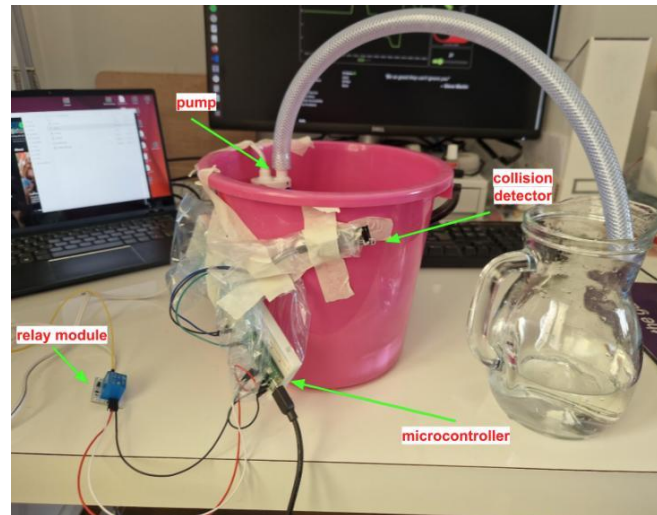


Simon Nyberg: Dog Activity Monitor

<https://hackmd.io/@Inu-iot/good-examples>



Hannes Månsson: Monitor plant environment



Elin Sigrid Margareta and Andersson Lundell: Smart Sangria Pump



Boris Smeds: Water Well Monitoring

Previous projects with SME – Astrid Lindgren's World

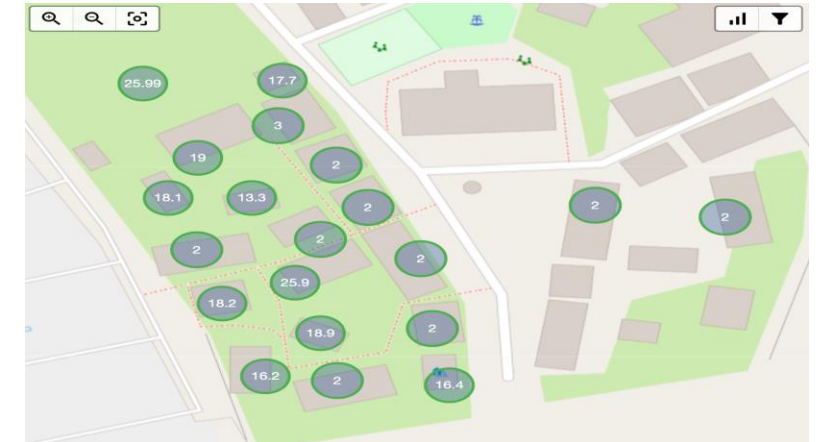
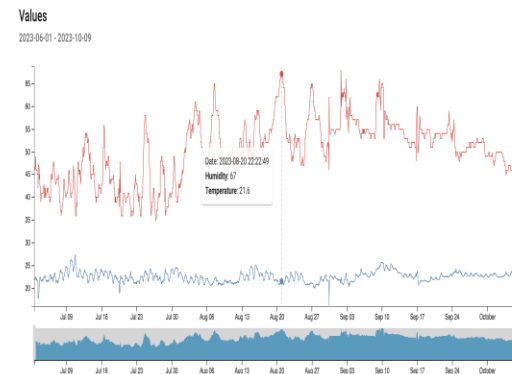
Objectives of the project:

- Measuring temperature and humidity in cabins
- Store data
- Visualize the data
- Automatic ventilation management and heat regulation (in the future)



Result:

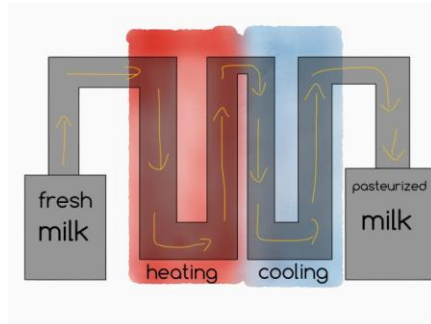
- ~35 temperature & humidity sensors deployed
- A visitor counter
- Dashboard for easier monitoring
- Alarm in case of abnormal values



Previous projects with SME - Öland's Köksmejeri

Objectives of the project:

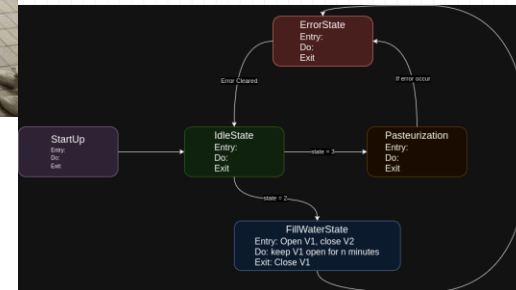
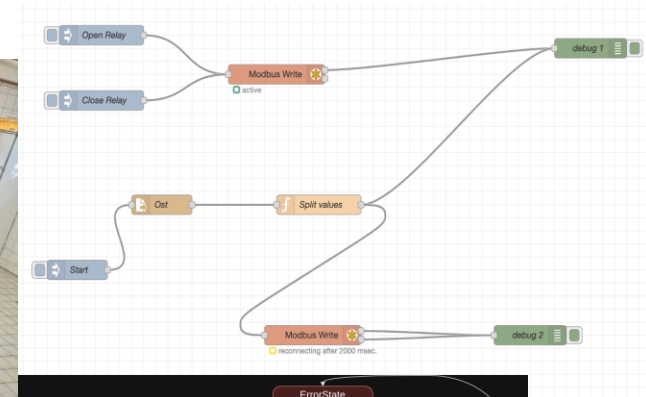
- Automate the pasteurization process
- Program recipes for different cheeses
- Alarm in case of abnormal values (temperature, time, pH?)
- Send log data to database (traceability)



ÖLANDS KÖKSMEJERI

Result:

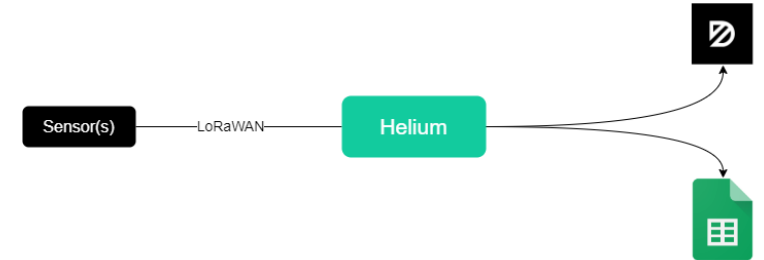
- Proof of Concept consisting of:
 - Industrial Shields PLC (esp32-38AR)
 - Flow meter
 - VFD (motor drive)
 - Logging throughout the pasteurization process



Previous projects with SME - Kallskänken (restaurant)

Objective of the project:

- Measuring of:
 - Number of visits
 - Noise level
 - Temperature and humidity
- Store data
- Visualize the data
- Download data to google sheet



Kallskänken®



Referred Books

1. Chou, Timothy. *Precision: principles, practices and solutions for the internet of things*. Lulu. com, 2016.
2. Slama, Dirk, Frank Puhlmann, Jim Morrish, and Rishi M. Bhatnagar. *Enterprise IoT: Strategies and Best practices for connected products and services*. " O'Reilly Media, Inc.", 2015.
3. Pfister, Cuno. *Getting started with the Internet of things: connecting sensors and microcontrollers to the cloud*. " O'Reilly Media, Inc.", 2011.
4. Verma, Jitendra Kumar, Deepak Kumar Saxena, Vicente González-Prida Díaz, and Vira Shendryk, eds. *Cloud IoT: Concepts, Paradigms, and Applications*. CRC Press, 2022.





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